Siemens S7-1200

CPU 1212C AC/DC/Relay

Word Logic Operations

- AND
- OR
- XOR
- INV
- DECODE
- ENCODE
- SELECT
- MUX
- DEMUX



Learning Made Easy

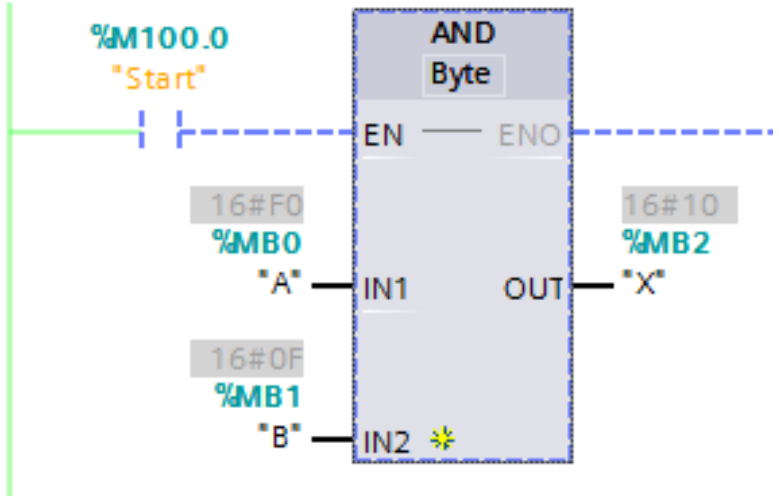
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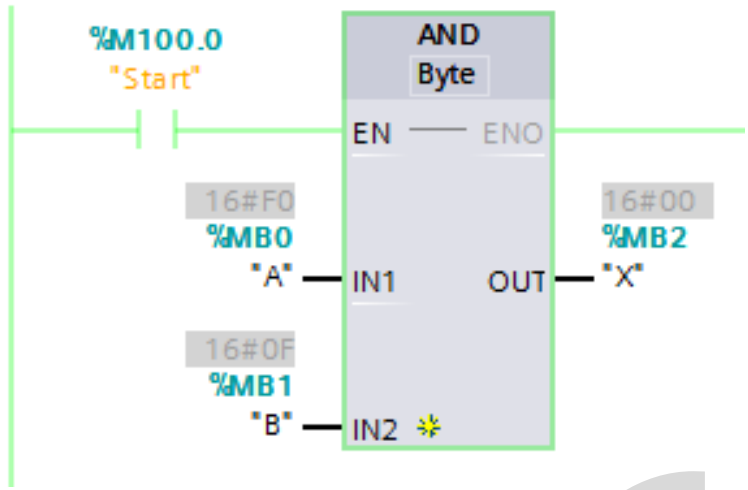


Understanding Word Operations – AND

AND Operation

You can use the "**AND logic operation**" instruction to combine the value at the IN1 input and the value at the IN2 input **bit-by-bit by AND logic** and query the result at the OUT output





How it works?

A	1	1	1	1	0	0	0	0
B	0	0	0	0	1	1	1	1
X	0	0	0	0	0	0	0	0

Bit wise AND Operation

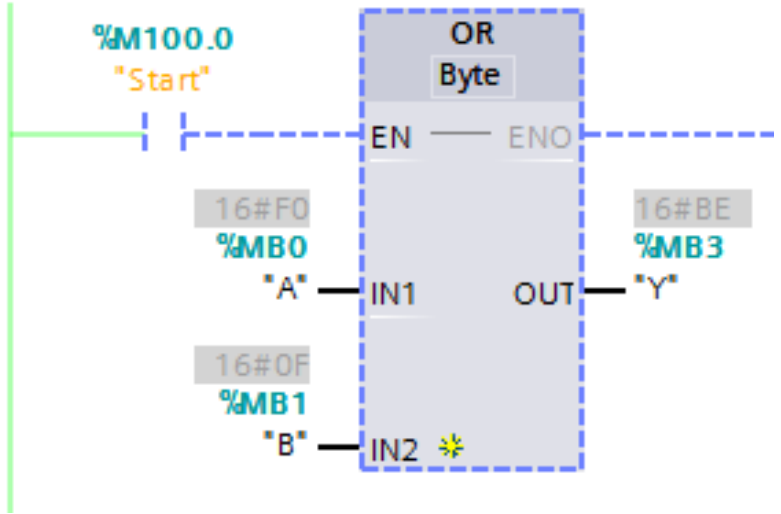
0 . 0 = 0
0 . 1 = 0
1 . 0 = 0
1 . 1 = 1

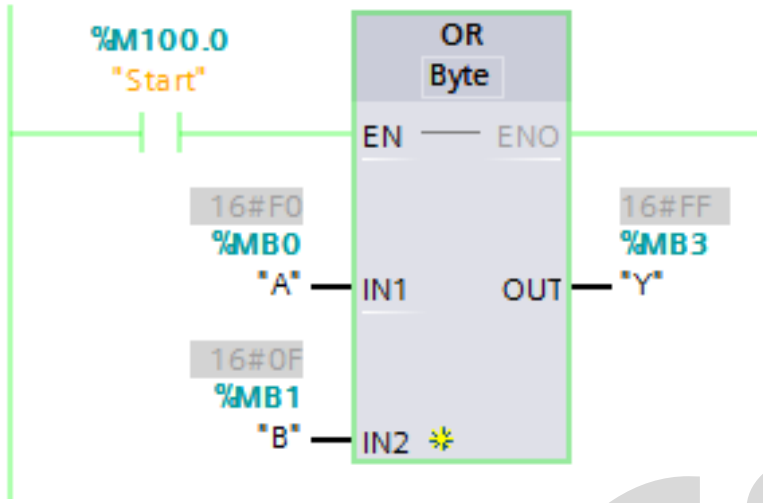


Understanding Word Operations – **OR**

OR Operation

You can use the "OR logic operation" instruction to combine the value at the IN1 input and the value at the IN2 input bit-by-bit by OR logic and query the result at the OUT output.





How it works?

A	1	1	1	1	0	0	0	0
B	0	0	0	0	1	1	1	1
Y	1	1	1	1	1	1	1	1

Bit wise OR Operation

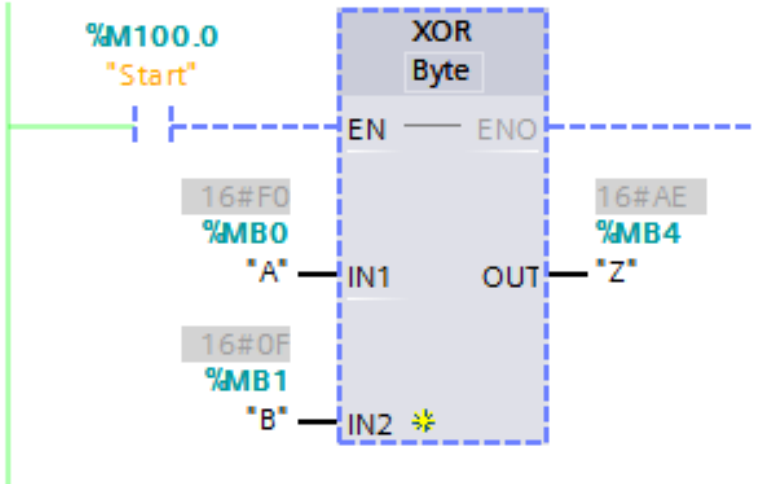
0 . 0 = 0
0 . 1 = 1
1 . 0 = 1
1 . 1 = 1

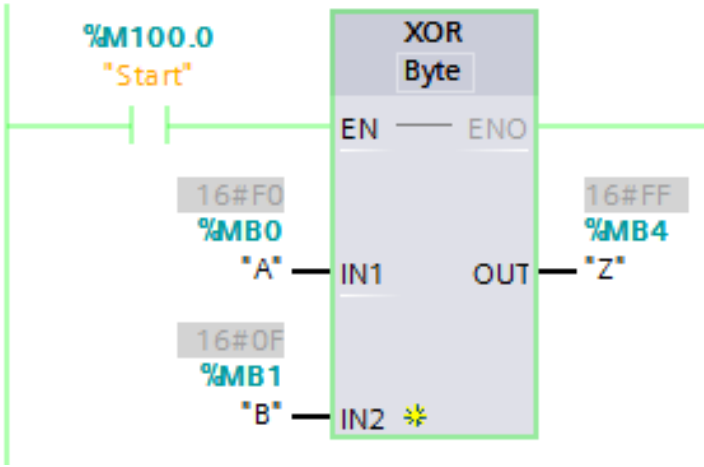


Understanding Word Operations – XOR

XOR Operation

You can use the "EXCLUSIVE OR logic operation" instruction to combine the value at the IN1 input and the value at the IN2 input bit-by-bit by EXCLUSIVE OR logic and query the result at the OUT output.





How it works?

A	1	1	1	1	0	0	0	0
B	0	0	0	0	1	1	1	1
Z	1	1	1	1	1	1	1	1

Bit wise XOR Operation

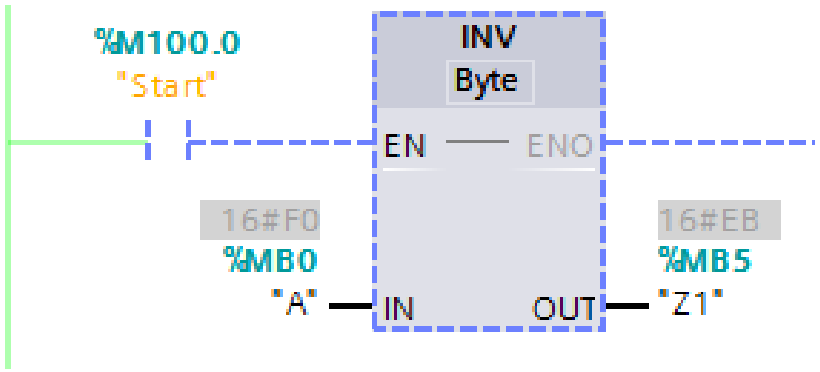
0 . 0 = 0
0 . 1 = 1
1 . 0 = 1
1 . 1 = 0

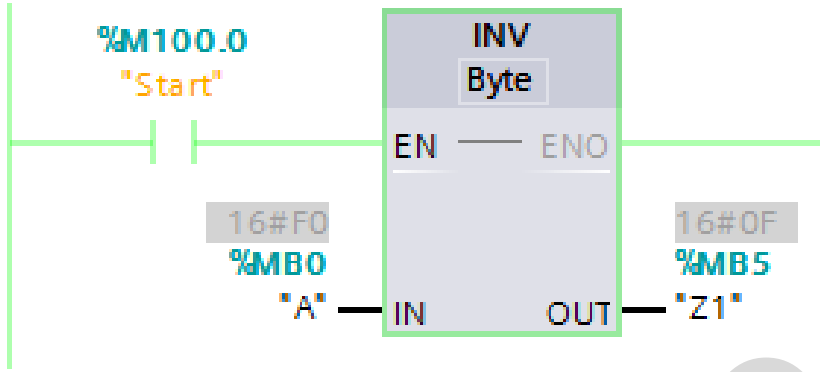


Understanding Word Operations – INV

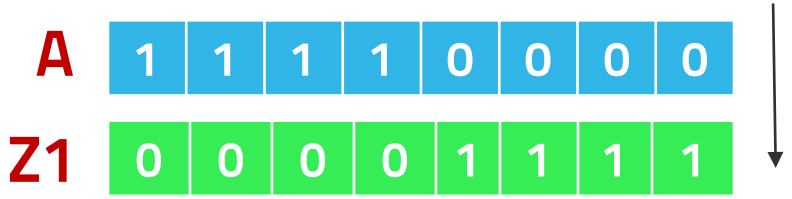
INV Operation

You can use the instruction "**Create ones complement**" to invert the signal status of the bits at input IN. This inverts the signal state of the individual bits that are then stored at output OUT.





How it works?



Inverse Operation

0 = 1

1 = 0

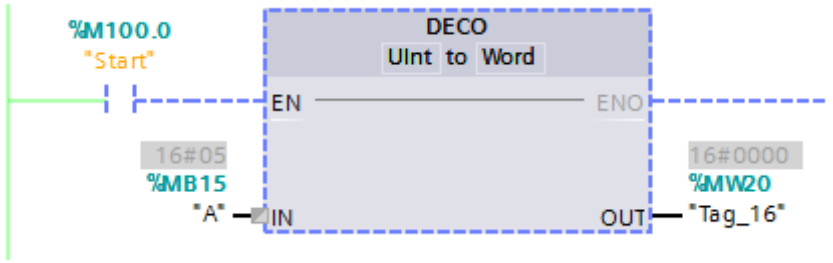


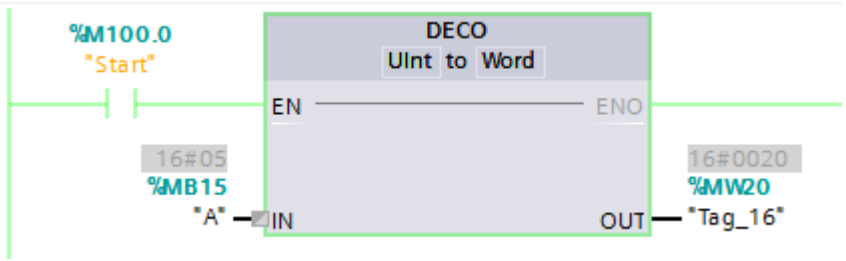
Understanding Word Operations – DECODE

DECODE Operation

You can use the "**Decode**" instruction to set a bit in the output value that is specified by the input value.

The "Decode" instruction reads the value at the IN input and **sets the bit in the output value** whose bit position corresponds to the read value. **The other bits in the output value are filled with zeroes.**





How it works?

IN 5

15.. ...0

OUT 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0

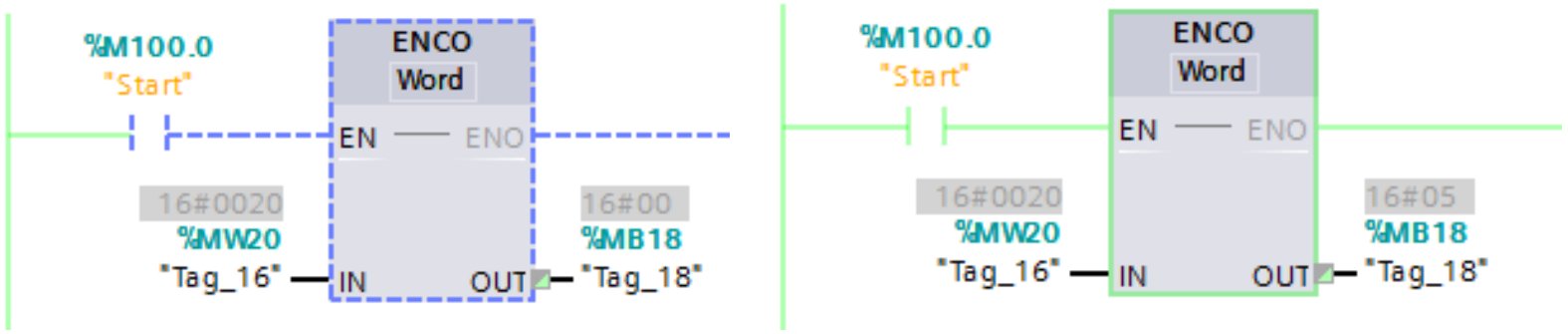
HEX = 16#0020



Understanding Word Operations – **ENCODE**

ENCODE Operation

You can use the "**Encode**" instruction to read the bit number of the least significant bit in the input value and to send it to the OUT output.
The "**Encode**" instruction selects the least significant bit of the value at the IN input and writes its bit number to the tag in the OUT output



How it works?

	15..															...0
IN	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
OUT	5															

HEX
= 16#0020



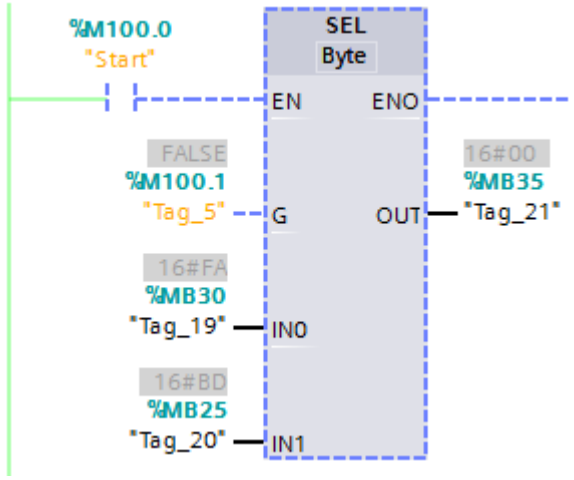
Understanding Word Operations – SELECT

SELECT Operation

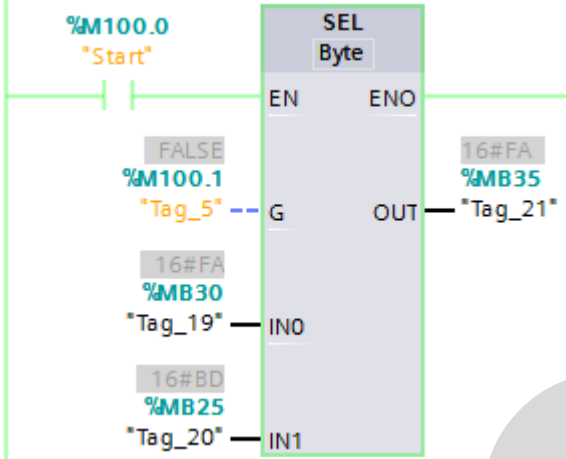
Depending on a **switch (G input),** the "**Select**" instruction selects one of the inputs, **IN0 or IN1** and copies its content to the OUT output.

All tags at all parameters must have the same data type.

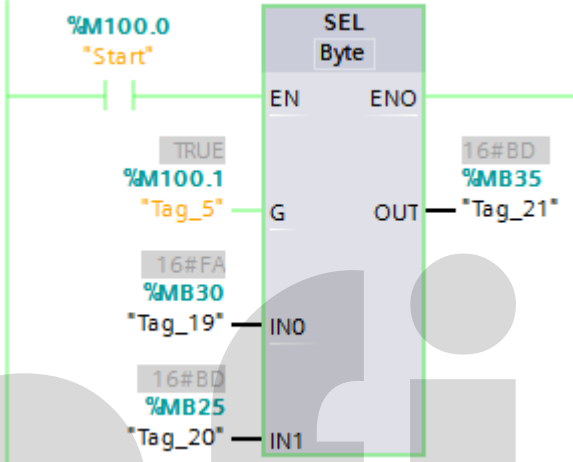
How it works?



EN = 0 | G = False
OUT = Last State



EN = 1 | G = False
OUT = IN0



EN = 1 | G = True
OUT = IN1

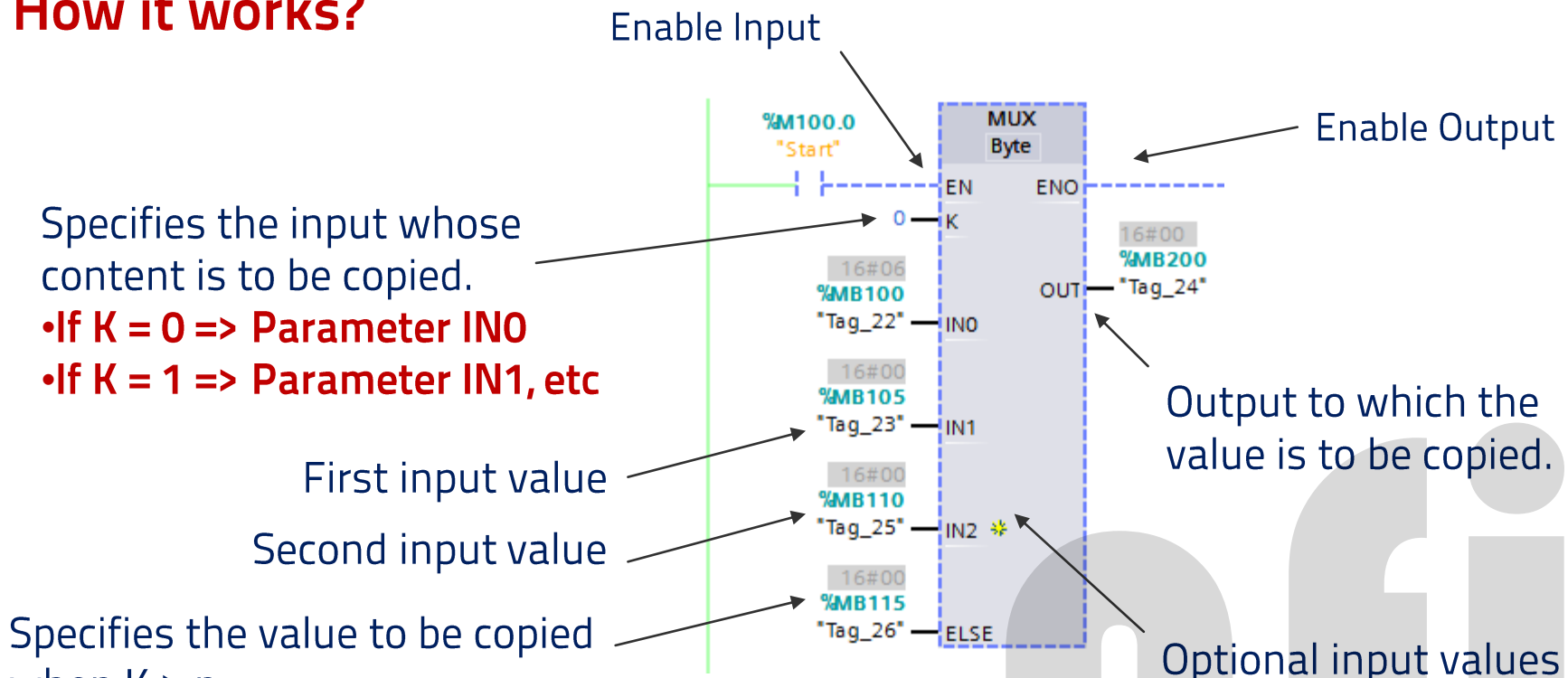


Understanding Word Operations – **MUX**

MUX Operation

You can use the instruction "**Multiplex**" to **copy the content of a selected input** to **output OUT**. The number of selectable inputs of the instruction box can be expanded. *You can declare a maximum of 32 inputs.*

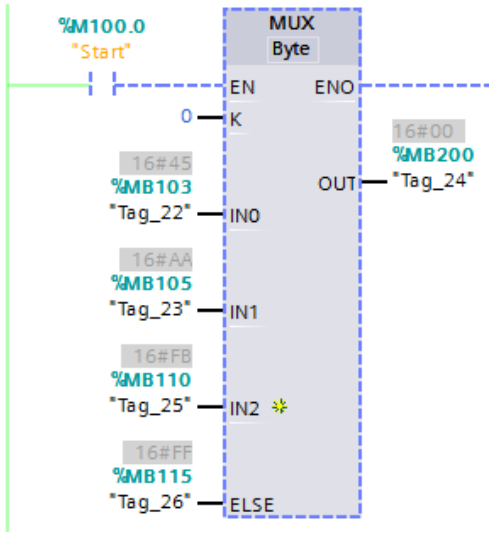
How it works?



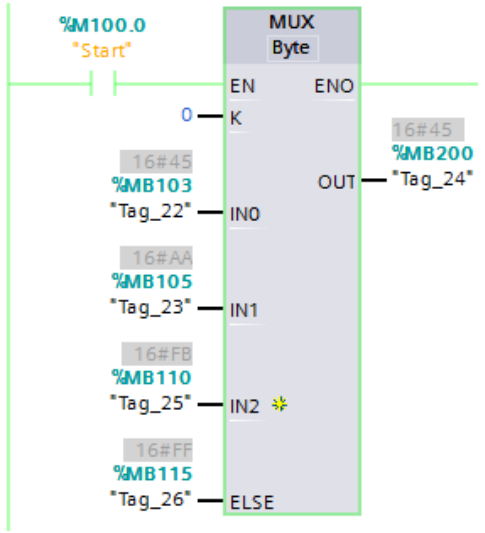


Understanding Word Operations – **MUX - Example**

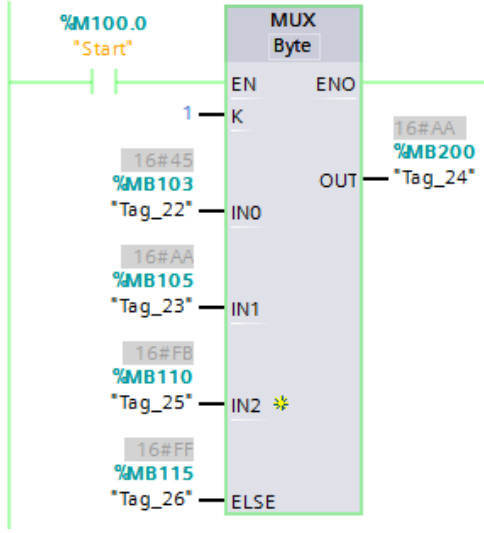
EN = 0 | K = 0 | OUT = Last State



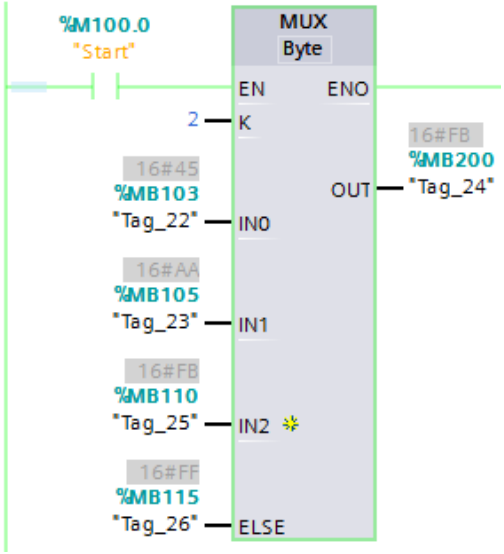
EN = 1 | K = 0 | OUT = IN0



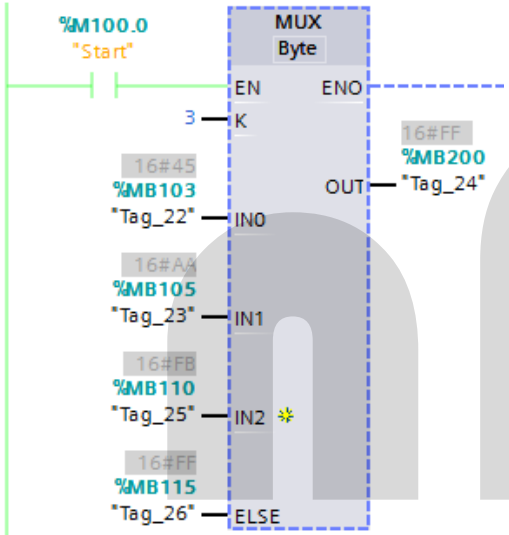
EN = 1 | K = 1 | OUT = IN1



EN = 1 | K = 2 | OUT = IN2



EN = 1 | K = 3 | OUT = ELSE



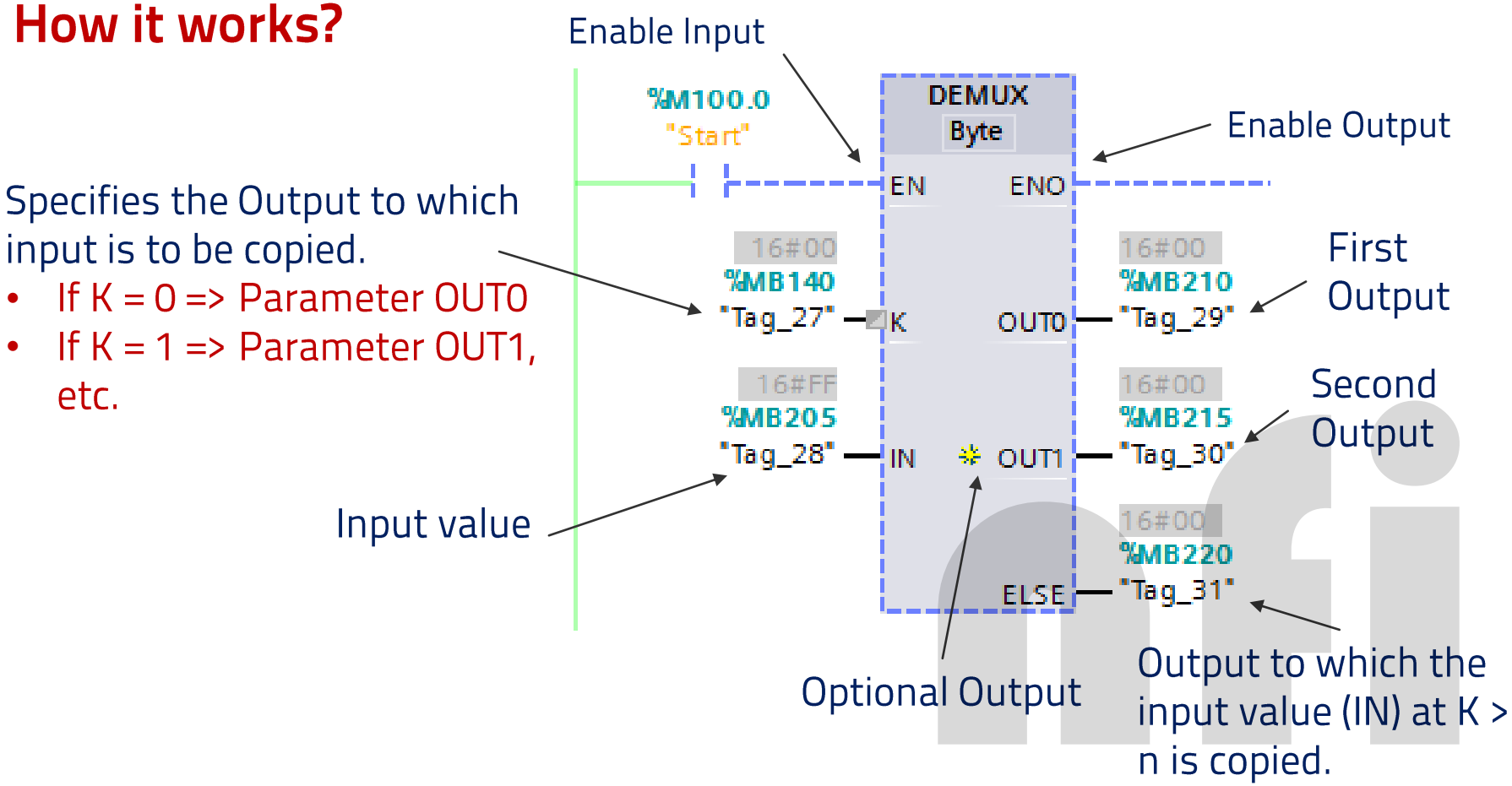


Understanding Word Operations – DEMUX

DEMUX Operation

You can use the instruction "**Demultiplex**" to **copy the content of the input IN to a selected output**. The number of selectable outputs can be extended in the instruction box. *Numbering starts at OUT0 and continues consecutively with each new output*

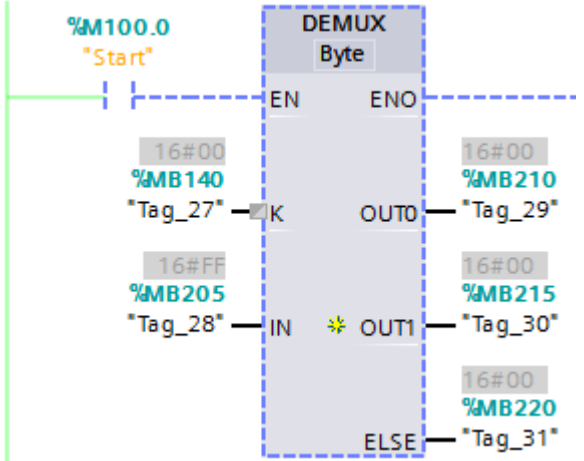
How it works?



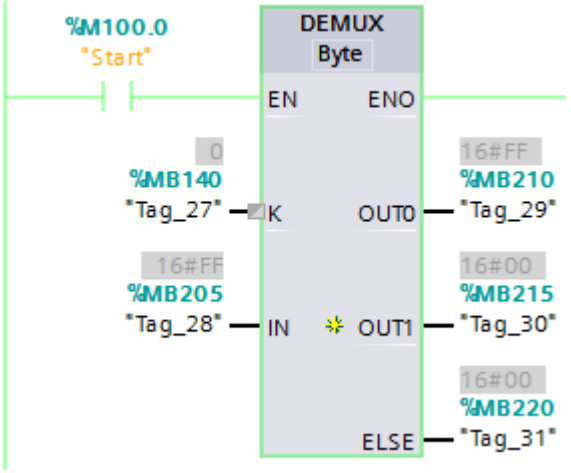


Understanding Word Operations – **DEMUX - Example**

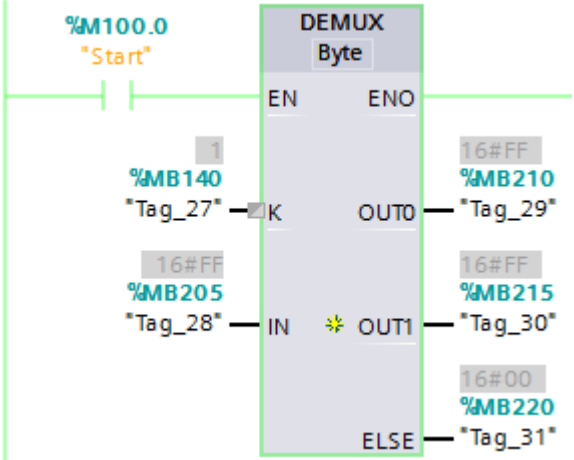
EN = 0 | K = 0
OUT 0 & 1 = Last State



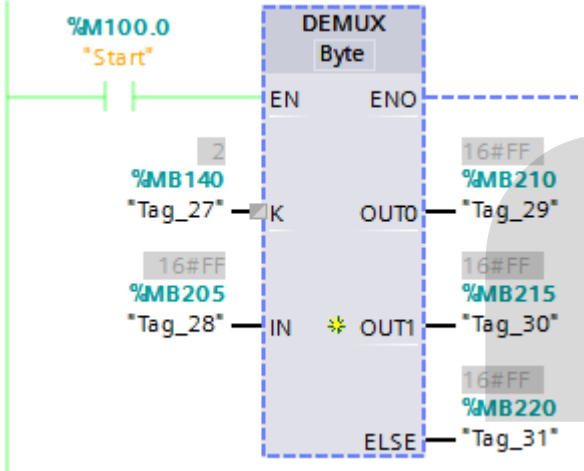
EN = 1 | K = 0
OUT0 = IN



EN = 1 | K = 1
OUT1 = IN



EN = 1 | K = 2
ELSE = IN



What did we learn today?

- **AND**: bitwise AND operation where output is true if both the inputs are TRUE
- **OR:** bitwise OR operation where output is true if any of both the input are TRUE
- **XOR**: bitwise OR operation where output is true if any one of the two inputs is TRUE
- **INV:** used for 1's complement of the input
- **DECODE:** instruction to set a bit in the output value that is specified by the input value.
- **ENCODE:** You can use the "Encode" instruction to read the bit number of the least significant bit in the input value and to send it to the OUT output
- **SELECT:** Depending on a switch (G input), the "Select" instruction selects one of the inputs, IN0 or IN1 and copies its content to the OUT output
- **MUX:** You can use the instruction "Multiplex" to copy the content of a selected input to output OUT
- **DEMUX:** You can use the instruction "Demultiplex" to copy the content of the input IN to a selected output.

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Thank you

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